Machine Learning: Introduction

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M2-OSAE 2022

Planning (adaptable)

Lesson 1 : Introduction to Machine Learning and simple algorithms hands-on

Lesson 2 – 3 : Artificial Neural Networks presentation, simple guided implementation, Multi-Layer Perceptron principles and usage

Lesson 4 – 5 : Advanced neural networks architectures and frameworks hands-on, application to more concrete cases (including astronomical data)

Exam
Duration: up to 3h.
Content: - 1h, Written-only part: theoretical notions from the lessons
- 2h, On-computer part: argumented network optimization problem

- Programming language poll: C, Fortran, Python ?
- Programming proficiency poll

Lessons materials

Slides, exercises, codes, corrections and datasets are available on GitHub and will be updated regularly:

http://github.com/Deyht/ML_OSAE_M2

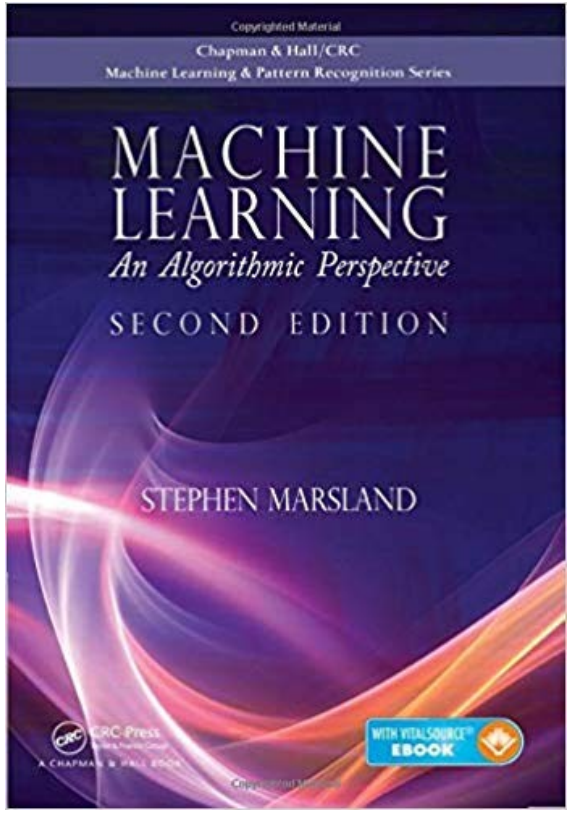
Useful commands:

git clone https://github.com/Deyht/Deyht/ML_OSAE_M2

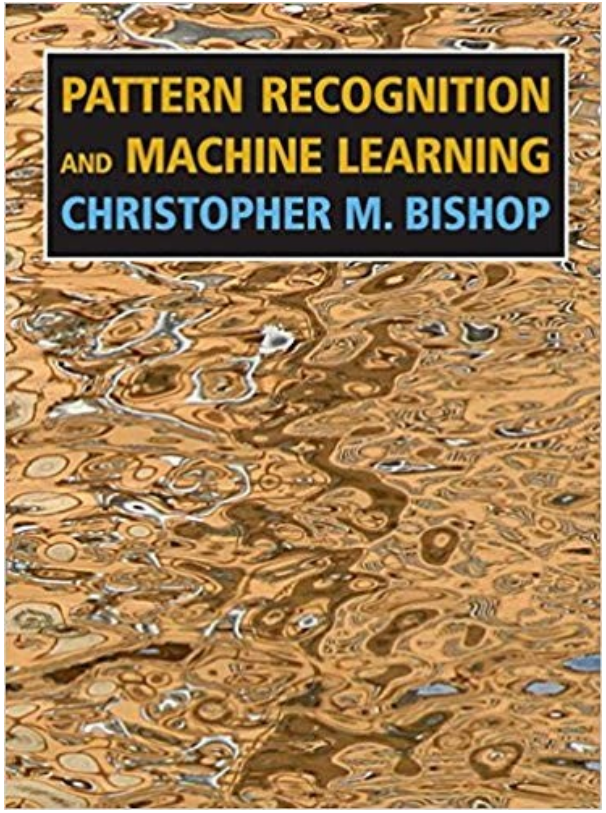
git pull

To avoid loosing your work on forced pull update, copy all files from the clone repository into a working directory!

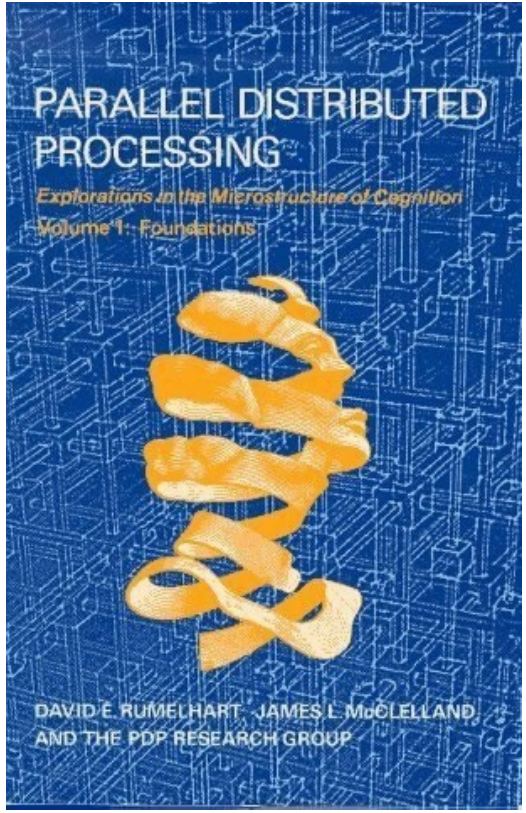
Some references



Stephen Marsland.
Machine Learning an Algorithmic Perspective.
Chapman and Hall/Crc, 2015



Christopher M. Bishop.
Pattern Recognition and Machine Learning.
Springer, 2011



David E. Rumelhart et al.
Parallel Distributed Processing. MIT press, 1989

Some references

- My PhD Thesis: https://arxiv.org/abs/2010.01431
  Chapters 2, 4 and 11 can be considered as a handout for this course

- The Fidle (Formation Introduction au Deep Learning) course from the CNRS:
  https://gricad-gitlab.univ-grenoble-alpes.fr/talks/fidle/

- The Neural Network CS231n course from Stanford:
  https://cs231n.github.io/

- Some more references on a dedicated page on the MINERVA website
  https://minerva-astro.obspm.fr/learning/

Take everything you read about ML and AI with a grain of salt ! (This course included)

Lots of materials still contains practical or theoretical approximations and errors.
Crossing references is most of the time necessary.